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Electronic package and manufacturing method thereof

Abstract

An electronic package, in which a heat dissipation structure is disposed on a carrier structure to form a packaging space for electronic components to be accommodated in the packaging space, and the electronic components are completely encapsulated by a heat dissipation material to prevent the electronic components exposing from the heat dissipation material so as to improve the heat dissipation effect.

Inventors: Huang; Yu-Lung (Taichung, TW), Huang; Chih-Ming (Taichung, TW), Yu; Kuo-Hua (Taichung, TW), Lin; Chang-Fu (Taichung, TW)

Applicant: SILICONWARE PRECISION INDUSTRIES CO., LTD. (Taichung, TW)

Family ID: 1000008781583

Assignee: SILICONWARE PRECISION INDUSTRIES CO., LTD. (Taichung, TW)

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Primary Examiner: Rosario; Joshua Benitez

Assistant Examiner: Nelson; Jacob T

Attorney, Agent or Firm: Dority & Manning, PA

Background/Summary

BACKGROUND

1. Technical Field

(1) The present disclosure relates to a semiconductor packaging process, and more particularly, to an electronic package with a heat dissipation structure and a manufacturing method thereof.

2. Description of Related Art

(2) With the improvements of requirements in functions and process speed for electronic products, a semiconductor chip as a core component of the electronic product requires electronic components and electronic circuits with higher density, so a greater amount of heat will be generated from the semiconductor chip during its operation. Besides, since the conventional packaging colloid for encapsulating the semiconductor chip is a poor heat transfer material (i.e., the efficiency of the heat dissipation is poor) with thermal conductivity of only 0.8 Watt/meter-Kelvin (W.Math.m.sup.-1.Math.k.sup.-1), damages of the semiconductor chips and product reliability problems will be occurred if the heat generated from the semiconductor chips cannot be dissipated effectively.

(3) Therefore, in order to rapidly dissipate the heat to the outside, a heat sink (heat spreader) is usually disposed in the semiconductor packages, the heat sink is bonded onto the backside of the semiconductor chip by a heat dissipation colloid, such as a thermal interface material (TIM), so as to dissipate the heat generated from the semiconductor chip by the heat dissipation colloid and the heat sink; in addition, the top of the heat sink is usually exposed from the packaging colloid or directly exposed to the atmosphere to obtain a better heat dissipation effect.

(4) As shown in FIG. 1, in a conventional semiconductor package **1**, a semiconductor chip **11** is firstly disposed on a packaging substrate **10** with its active surface **11a** by flip-chip bonding (i.e., through conductive bumps **110** and an underfill **111**), and then a heat dissipation member **13** is

bonded onto the inactive surface **11b** of the semiconductor chip **11** with its top sheet **130** by a TIM layer **12**, and supporting legs **131** of the heat dissipation member **13** are disposed on the packaging substrate **10** via an adhesive layer **14**.

(5) During the operation, the heat generated from the semiconductor chip **11** is transferred to the top sheet **130** of the heat dissipation member **13** by the inactive surface **11b** and the TIM layer **12** so as to be dissipated to outside of the semiconductor package **1**.

(6) However, in the conventional semiconductor package **1**, the TIM layer **12** is only laid on the inactive surface **11b** of the semiconductor chip **11**, such that the heat dissipation effect is poor.

(7) Therefore, how to overcome the aforementioned drawbacks of the prior art has become an urgent issue to be solved at present.

SUMMARY

(8) In view of the aforementioned shortcomings of the prior art, the present disclosure provides an electronic package, which comprises: a carrier structure; an electronic component disposed on the carrier structure; a heat dissipation structure including a heat dissipation body covering the electronic component and supporting legs vertically disposed on the heat dissipation body, wherein the heat dissipation structure is bonded onto the carrier structure via the supporting legs, such that a packaging space is formed between the heat dissipation body and the carrier structure, and the electronic component is disposed in the packaging space; and a heat dissipation material disposed in the packaging space and completely encapsulating the electronic component, wherein the electronic component is free from being exposed from the heat dissipation material.

(9) The present disclosure further provides a manufacturing method for an electronic package, which comprises: providing a heat dissipation structure and a carrier structure with an electronic component, wherein the heat dissipation structure includes a heat dissipation body covering the electronic component and supporting legs vertically disposed on the heat dissipation body; bonding the supporting legs to the carrier structure to form a packaging space between the heat dissipation body and the carrier structure, wherein the electronic component is located in the packaging space; and disposing a heat dissipation material in the packaging space to completely encapsulate the electronic component, wherein the electronic component is free from being exposed from the heat dissipation material.

(10) In the aforementioned electronic package and manufacturing method thereof, the electronic component is in a form of laid package.

(11) In the aforementioned electronic package and manufacturing method thereof, the electronic component is in a form of upright package.

(12) In the aforementioned electronic package and manufacturing method thereof, the supporting legs are fixed on the carrier structure via a bonding material. For instance, the supporting legs are formed with accommodation ports for filling the bonding material.

(13) In the aforementioned electronic package and manufacturing method thereof, the supporting legs and the carrier structure are spaced apart to form accommodation spaces. For instance, the accommodation spaces are accommodated with passive components.

(14) In the aforementioned electronic package and manufacturing method thereof, the packaging space is free from being filled up with the heat dissipation material, and remaining spaces of the packaging space are used as target spaces.

(15) In the aforementioned electronic package and manufacturing method thereof, the heat dissipation structure is formed with a first port and a second port that are communicating the packaging space, and the first port and the second port are located at different locations, wherein the heat dissipation material is filled into the packaging space via the first port, and the second port is used to exhaust air. For instance, the present disclosure further comprises covering the first port and the second port respectively via a plurality of plugs after the heat dissipation material completely encapsulating the electronic component.

(16) In the aforementioned electronic package and manufacturing method thereof, the supporting

legs cover side surfaces of the carrier structure.

(17) In the aforementioned electronic package and manufacturing method thereof, the present disclosure further comprises clamping the supporting legs and the carrier structure via clamps.

(18) As can be understood from the above, in the electronic package and manufacturing method thereof of the present disclosure, the electronic component is not exposed from the heat dissipation material since the electronic component is completely encapsulated by the heat dissipation material. Thus, compared with the prior art, the present disclosure is capable of substantially improving the heat dissipation effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic cross-sectional view of a conventional semiconductor package.

(2) FIG. 2A-1 is a schematic cross-sectional view of an electronic package according to a first embodiment of the present disclosure.

(3) FIG. 2A-2 and FIG. 2A-3 are schematic partial cross-sectional views of different aspects of FIG. 2A-1.

(4) FIG. 2B is a schematic top view along a cross section A-A of FIG. 2A-1.

(5) FIG. 2C is a schematic cross-sectional view along a cross-section line B-B of FIG. 2B.

(6) FIG. 2D is a schematic cross-sectional view along a cross-section line C-C of FIG. 2B.

(7) FIG. 3A to FIG. 3E are schematic cross-sectional views illustrating a manufacturing method of an electronic package according to a second embodiment of the present disclosure.

(8) FIG. 4A is a schematic cross-sectional view of an electronic package according to the second embodiment of the present disclosure.

(9) FIG. 4B is a schematic top view along a cross-section D-D of FIG. 4A.

(10) FIG. 4C is a schematic cross-sectional view along a cross-section line E-E of FIG. 4B.

(11) FIG. 4D is a schematic cross-sectional view along a cross-section line F-F of FIG. 4B.

(12) FIG. 5A-1 is a schematic top view of an electronic package according to a third embodiment of the present disclosure.

(13) FIG. 5A-2 is a schematic top view of another aspect of FIG. 5A-1.

(14) FIG. 5B is a schematic cross-sectional view along a cross-section line G-G of FIG. 5A-1.

(15) FIG. 5C is a schematic cross-sectional view along a cross-section line H-H of FIG. 5A-1.

(16) FIG. 5D is a schematic cross-sectional view along a cross-section line I-I of FIG. 5A-1.

(17) FIG. 5E is a schematic cross-sectional view along a cross-section line J-J of FIG. 5D.

(18) FIG. 5F-1 is a schematic cross-sectional view of another aspect of FIG. 5B.

(19) FIG. 5F-2 is a schematic cross-sectional view of another aspect of FIG. 5F-1.

DETAILED DESCRIPTION

(20) Implementations of the present disclosure are illustrated using the following embodiments.

One of ordinary skill in the art can readily appreciate other advantages and technical effects of the present disclosure upon reading the content of this specification.

(21) It should be noted that the structures, ratios, sizes shown in the drawings appended to this specification are to be construed in conjunction with the disclosure of this specification in order to facilitate understanding of those skilled in the art. They are not meant, in any ways, to limit the implementations of the present disclosure, and therefore have no substantial technical meaning. Any modifications, changes or adjustments to the structures, ratio relationships or sizes, are to be construed as falling within the range covered by the technical content disclosed herein to the extent of not causing changes in the technical effects created and the objectives achieved by the present disclosure. Meanwhile, terms such as “above,” “first,” “second,” “a,” “an,” and the like recited herein are for illustrative purposes, and are not meant to limit the scope in which the present

disclosure can be implemented. Any variations or modifications to their relative relationships, without changes in the substantial technical content, should also to be regarded as within the scope in which the present disclosure can be implemented.

(22) FIG. 2A-1 is a schematic cross-sectional view of an electronic package 2 according to a first embodiment of the present disclosure.

(23) As shown in FIG. 2A-1, the electronic package 2 comprises: a carrier structure 24, at least one electronic component 25 disposed on the carrier structure 24, a heat dissipation structure 2a disposed on the carrier structure 24 to cover the electronic component 25, and a heat dissipation material 23 for encapsulating the electronic component 25, wherein the heat dissipation structure 2a comprises a sheet-shaped heat dissipation body 20 and a plurality of first supporting legs 21 and a plurality of second supporting legs 22 vertically disposed on the heat dissipation body 20.

(24) A protruding heat dissipation block 200 is integrally formed with and on a partial surface of the heat dissipation body 20.

(25) In an embodiment, the heat dissipation block 200 is in a shape of a cone or a hill, and a top end of the heat dissipation block 200 is closest to the electronic component 25.

(26) The first supporting leg 21 is used as a wall-shaped outer leg, as shown in FIG. 2B, the first supporting leg 21 is vertically disposed on the carrier structure 24 by a bonding material 26 such as a glue material.

(27) In an embodiment, the first supporting leg 21 can be formed with at least one accommodation port 210 that can be filled with a colloid (such as the bonding material 26) in accordance to requirements.

(28) The second supporting leg 22 is used as a column-shaped or wall-shaped inner leg, as shown in FIG. 2B, the second supporting leg 22 and the carrier structure 24 are spaced apart to form an accommodation space S1 for accommodating at least one passive component 27 (such as a resistor, a capacitor, and an inductor) on the carrier structure 24, wherein the passive component 27 is electrically connected to circuit layers of the carrier structure 24 and is encapsulated by an insulating material 270.

(29) In another embodiment, as shown in FIG. 2A-2, the second supporting leg 22 can be formed with at least one accommodation port 220 that can be coated with a colloid (such as the bonding material 26) to engage with a protruding portion 240 of the carrier structure 24 according to requirements, thereby neglecting the configuration of the first supporting leg 21, wherein the passive component 27 can be disposed on the carrier structure 24 at the periphery of the second supporting leg 22. Alternatively, as shown in FIG. 2A-3, the first supporting leg 21 and the second supporting leg 22 are both vertically disposed on the carrier structure 24 to form another accommodation space S2 between the first supporting leg 21 and the second supporting leg 22 for disposing at least one passive component 27 in the accommodation space S2 on the carrier structure 24, wherein the passive component 27 is electrically connected to circuit layers of the carrier structure 24.

(30) As a result, the heat dissipation body 20 is disposed on the carrier structure 24 by the first supporting leg 21 and/or the second supporting leg 22, so that a packaging space S (which may include the accommodation spaces S1, S2 in accordance to requirements) is formed between the heat dissipation body 20 and the carrier structure 24 for accommodating the electronic component 25 (and the passive component 27) in the packaging space S (and the accommodation spaces S1, S2).

(31) The heat dissipation material 23 is formed in the packaging space S to encapsulate the electronic component 25, and the packaging space S is not filled up with the heat dissipation material 23, so that the remaining spaces of the packaging space S are used as target spaces A1, as shown in FIG. 2A-1, the target spaces A1 are formed between the heat dissipation material 23 and the heat dissipation body 20.

(32) In an embodiment, the heat dissipation material 23 has a high thermal conductivity, about 30-

80 Watt/m-Kelvin (Wm.sup.-1K.sup.-1). For example, when an underfill (not shown) is formed between an active surface **25a** of the electronic component **25** and the carrier structure **24** to insulate conductive bumps **250** from the outside, then the heat dissipation material **23** is solid indium, liquid metal or any other metal-containing material that is fluid at room temperature/high temperature to be used as a thermal interface material (TIM), such as a low-temperature melting thermally conductive material. If there is no underfill (not shown) between the active surface **25a** of the electronic component **25** and the carrier structure **24**, then liquid oil or other liquids without metal material can be used as a heat dissipation material.

(33) Moreover, during the manufacturing process, the heat dissipation material **23** is formed by a filling method. For example, a first port **201** and a second port **202** are formed between the heat dissipation body **20** and the first supporting leg **21** (or the second supporting leg **22** shown in FIG. 2A-2) at different locations, as shown in FIG. 2A-1, the first port **201** and the second port **202** communicate the packaging space S, so that after the heat dissipation structure **2a** is fixed on the carrier structure **24**, the heat dissipation material **23** is injected into the packaging space S via the first port **201** acting as a filling port, such that the heat dissipation material **23** completely encapsulates the electronic component **25**, and after filling with the heat dissipation material **23**, the first port **201** is covered by a plug **28**, as shown in FIG. 2A-1 and FIG. 2B. It should be understood that the second port **202** is used as an exhausting port to facilitate the flow of the heat dissipation material **23** to discharge a large amount of air out of the packaging space S, and after filling with the heat dissipation material **23**, the second port **202** is covered by the plug **28**, as shown in FIG. 2A-1 and FIG. 2B, so that a portion of the air remains in the target space A1.

(34) The carrier structure **24** is a rectangular body, such as a packaging substrate with a core layer and a circuit structure, a packaging substrate with a coreless circuit structure, through silicon interposer (TSI) with a conductive through-silicon via (TSV) or other plate types, the carrier structure **24** comprises at least one insulating layer and at least one circuit layer bonded with the insulating layer, such as at least one fan-out type redistribution layer (RDL). It can be understood that the carrier structure **24** can also be other plates for carrying chips, such as lead frames, wafers, or other plate bodies with metal routings, etc., which are not limited to the above.

(35) In an embodiment, the carrier structure **24** is placed on a process machine, such that the first port **201** and the second port **202** of the heat dissipation structure **2a** face the left and right directions.

(36) The electronic component **25** is an active component, a passive component, a chip module, or a combination thereof, wherein the active component is for example a semiconductor chip, and the passive component is for example a resistor, a capacitor, and an inductor.

(37) In an embodiment, the electronic component **25** is in a form of laid package semiconductor chip, which has an active surface **25a** (or bottom surface) and an inactive surface **25b** (or top surface) opposing to the active surface **25a**, such that the active surface **25a** is disposed on the circuit layers of the carrier structure **24** in a flip-chip manner via a plurality of conductive bumps **250** such as solder materials, metal pillars, or others, and the electronic component **25** is electrically connected to the circuit layers, and the heat dissipation material **23** or underfill (not shown) encapsulates each of the conductive bumps **250**; alternatively, the electronic component **25** can be electrically connected to the circuit layers of the carrier structure **24** via a plurality of bonding wires (not shown) in a wire-bonding manner; or the electronic component **25** can directly contact the circuit layers of the carrier structure **24**. It should be understood that there are various means for the electronic component **25** to be electrically connected to the carrier structure **24**, and the required types and quantities of the electronic components can be disposed on the carrier structure **24** and not limited to the above.

(38) Hence, in the electronic package **2** of the present disclosure, the surface height H1 (i.e., the liquid height position) of the heat dissipation material **23** relative to the carrier structure **24** is higher than the surface height H2 (i.e., the height position of the inactive surface **25b**) of the

electronic component **25** relative to the carrier structure **24**, such that the heat dissipation material **23** completely encapsulates the active surface **25a**, the inactive surface **25b**, and the side surfaces **25c** of the electronic component **25** (as shown in FIG. 2C and FIG. 2D, which are schematic cross-sectional views along cross-section lines B-B and C-C of FIG. 2B), so the electronic component **25** is not exposed from the heat dissipation material **23**, therefore the electronic component **25** can effectively dissipate heat via the heat dissipation material **23**.

(39) Moreover, for the heat source of the electronic component **25**, the heat dissipation structure **2a** is closer to the heat source by the design of the heat dissipation block **200** to enhance the heat dissipation effect, and the heat dissipation block **200** is embedded in the heat dissipation material **23** to further enhance the heat dissipation effect.

(40) Furthermore, the first supporting leg **21** is wall-shaped to prevent the liquid heat dissipation material **23** from overflowing.

(41) In addition, the intervals between the first supporting leg **21** and the second supporting leg **22** are used as channels, as shown in FIG. 2B, to accommodate the liquid heat dissipation material **23**, and since the heat source is a specific area (i.e., the electronic component **25**), the liquid heat dissipation material **23** can be induced to generate convection activities.

(42) FIG. 3A to FIG. 3E are schematic cross-sectional views illustrating a manufacturing method of an electronic package **3** according to a second embodiment of the present disclosure. The difference between this embodiment and the first embodiment lies in the process configuration, and similarities between the two will not be provided to avoid repetitions.

(43) As shown in FIG. 3A, after the heat dissipation structure **2a** is fixed on the carrier structure **24**, it is rotated at 90 degrees to vertically dispose the heat dissipation structure **2a** and the carrier structure **24**, such that the first port **201** and the second port **202** of the heat dissipation structure **2a** face on top and bottom directions, so the electronic component **25** is in a form of upright package.

(44) As shown in FIG. 3B, the first port **201** located below the heat dissipation structure **2a** is used as a filling port, so that the heat dissipation material **33** is filled into the packaging space **S** from bottom to top by a filler **90**, and the second port **202** located over the heat dissipation structure **2a** is used as an exhausting port, so that an air nozzle **91** is disposed on the second port **202** for exhausting operation (e.g., the arrow direction **Y**).

(45) As shown in FIG. 3C, after the electronic component **25** is overflowed (or completely covered) by the heat dissipation material **33**, the air nozzle **91** is removed, and then the second port **202** is covered by the plug **28**.

(46) In an embodiment, the surface height **H2** (i.e., the liquid height position) of the heat dissipation material **33** relative to the first port **201** is higher than the surface height **H1** (i.e., the height position of the side surface **25c** farthest from the first port **201**) of the electronic component **25** relative to the first port **201**, such that the heat dissipation material **33** completely encapsulates the active surface **25a**, the inactive surface **25b**, and the side surfaces **25c** of the electronic component **25**, so that the electronic component **25** can be effectively dissipated heat by the heat dissipation material **33**.

(47) As shown in FIG. 3D, the electronic package **3** is turned upside down, so the first port **201** is located over the heat dissipation structure **2a** to be used as an exhausting port to dispose another air nozzle **81** on the first port **201** for exhausting operation (e.g., the arrow direction **Y**). The first port **201** is covered by another plug **28** after the air inside the packaging space **S** is discharged, so the target space **A2** is formed between the upper first supporting leg **21** and the heat dissipation material **23**.

(48) As shown in FIG. 3E, another air nozzle **81** is removed to form the electronic package **3**.

(49) As a result, in the electronic package **3** of the present disclosure, the surface height **H1** (i.e., the liquid height position shown in FIG. 4A and FIG. 4B) of the heat dissipation material **33** relative to the second port **202** is higher than the surface height **H2** (i.e., the height position of the side surface **25c** farthest away from the second port **202** shown in FIG. 4A and FIG. 4B) of the

electronic component **25** relative to the second port **202**, such that the heat dissipation material **33** completely encapsulates the active surface **25a**, the inactive surface **25b**, and the side surfaces **25c** of the electronic component **25** (as shown in FIG. 4A, FIG. 4C, FIG. 4D), so that the electronic component **25** is not exposed from the heat dissipation material **33**, and the electronic component **25** can effectively dissipate heat via the heat dissipation material **33**.

(50) FIG. 5A-1 and FIG. 5B are schematic cross-sectional views of an electronic package **5** according to a third embodiment of the present disclosure. The difference between this embodiment and the above embodiments lies in the design of the heat dissipation structure **5a**, and similarities between the embodiments will not be provided to avoid repetitions.

(51) As shown in FIG. 5B, in a form of laid package, the first supporting leg **51** of the heat dissipation structure **5a** is bonded to the side surface **24c** of the carrier structure **24** via the bonding material **26**, and a plurality of conductive components **59** containing solder materials can be arranged at the bottom of the carrier structure **24**. It should be understood that in a form of upright package (as shown in FIG. 5F-1), the first supporting leg **51** of the heat dissipation structure **5b** can also be bonded to the side surface **24c** of the carrier structure **24** via the bonding material **26**.

(52) In an embodiment, the first port **501** and the second port **502** are formed through the heat dissipation body **20**, as shown in FIG. 5A-1 and FIG. 5D.

(53) Furthermore, at least one detection recess **500** can be formed on the inner side of the heat dissipation body **20**, such as four as shown in FIG. 5A-1, for detecting whether the surface height (i.e., the liquid height position) of the heat dissipation material **23** relative to the carrier structure **24** is higher than the surface height (i.e., the height position of the inactive surface **25b**) of the electronic component **25** relative to the carrier structure **24**, thereby to effectively ensure that the heat dissipation material **23** completely encapsulates the electronic component **25**, as shown in FIG. 5C, FIG. 5D and FIG. 5E.

(54) Hence, when the detection instrument (not shown) detects a predetermined height (e.g., the detection recess **500** is filled up with the heat dissipation material **23** as shown in FIG. 5D), it can be determined that the heat dissipation material **23** has completely encapsulated the electronic component **25**.

(55) Moreover, the packaging space **S** will be presented in negative pressure after the exhausting operation, so the target spaces **A1**, **A2** are formed by the heat dissipation material **23** not filling up the packaging space **S**, so that the heat dissipation material **23** can be filled into the target spaces **A1**, **A2** when thermally expanded in the subsequent thermal process, thereby avoiding the problem of plate explosion.

(56) Further, as shown in FIG. 5A-2, in a form of laid package, after the first supporting leg **21** of the heat dissipation structure **2a** is bonded to the carrier structure **24** via the bonding material **26**, the first supporting leg **51** and the side surface **24c** of the carrier structure **24** can be clamped by clamps **60** for facilitating the subsequent process. For instance, the clamps **60** can be located outside the four edges of the heat dissipation body **20** respectively. It can be understood that in a form of upright package (as shown in FIG. 5F-2), the first supporting leg **21** and the side surface **24c** of the carrier structure **24** can also be clamped by the clamps **60**.

(57) To sum up, in the electronic package and the manufacturing method thereof of the present disclosure, the electronic component **25** is effectively dissipated heat mainly by the heat dissipation material **23**, **33** completely encapsulating the electronic component **25**, such that whether the electronic component **25** is in a form of laid package or upright package, both the heat dissipation materials **23**, **33** completely encapsulate the electronic component **25**.

(58) The above embodiments are set forth to illustrate the principles of the present disclosure, and should not be interpreted as to limit the present disclosure in any way. The above embodiments can be modified by one of ordinary skill in the art without departing from the scope of the present disclosure as defined in the appended claims. Therefore, the scope of protection of the right of the present disclosure should be listed as the following appended claims.

Claims

1. An electronic package, comprising: a carrier structure; an electronic component disposed on the carrier structure; a heat dissipation structure including a heat dissipation body covering the electronic component and supporting legs vertically disposed on the heat dissipation body, wherein the heat dissipation structure is bonded onto the carrier structure via the supporting legs, such that a packaging space is formed between the heat dissipation body and the carrier structure, and the electronic component is disposed in the packaging space, wherein the supporting legs are formed with accommodation ports coated with a bonding material to engage with protruding portions of the carrier structure; and a heat dissipation material disposed in the packaging space and completely encapsulating the electronic component, wherein the electronic component is free from being exposed from the heat dissipation material.
2. The electronic package of claim 1, wherein the electronic component is in a form of a laid package.
3. The electronic package of claim 1, wherein the electronic component is in a form of an upright package.
4. The electronic package of claim 1, wherein the supporting legs and the carrier structure are spaced apart to form accommodation spaces.
5. The electronic package of claim 4, wherein the accommodation spaces are accommodated with passive components.
6. The electronic package of claim 1, wherein the packaging space is free from being filled up with the heat dissipation material.
7. The electronic package of claim 1, wherein the heat dissipation structure is formed with a first port and a second port that are communicating the packaging space, and the first port and the second port are located at different locations.
8. The electronic package of claim 7, further comprising a plurality of plugs for covering the first port and the second port respectively.
9. The electronic package of claim 1, wherein the supporting legs cover side surfaces of the carrier structure.
10. The electronic package of claim 1, further comprising clamps for clamping the supporting legs and the carrier structure.
11. A method for manufacturing an electronic package, comprising: providing a heat dissipation structure and a carrier structure with an electronic component, wherein the heat dissipation structure includes a heat dissipation body covering the electronic component and supporting legs vertically disposed on the heat dissipation body; bonding the supporting legs to the carrier structure to form a packaging space between the heat dissipation body and the carrier structure, wherein the supporting legs are formed with accommodation ports coated with a bonding material to engage with protruding portions of the carrier structure, wherein the electronic component is located in the packaging space; and forming a heat dissipation material in the packaging space to completely encapsulate the electronic component, wherein the electronic component is free from being exposed from the heat dissipation material.
12. The method of claim 11, wherein the electronic component is in a form of a laid package.
13. The method of claim 11, wherein the electronic component is in a form of an upright package.
14. The method of claim 11, wherein the supporting legs and the carrier structure are spaced apart to form accommodation spaces.
15. The method of claim 14, wherein the accommodation spaces are accommodated with passive components.
16. The method of claim 11, wherein the packaging space is free from being filled up with the heat dissipation material.

17. The method of claim 11, wherein the heat dissipation structure is formed with a first port and a second port that are communicating the packaging space, and the first port and the second port are located at different locations, wherein the heat dissipation material is filled into the packaging space via the first port, and the second port is used to exhaust air.

18. The method of claim 17, further comprising covering the first port and the second port respectively via a plurality of plugs after the heat dissipation material completely encapsulating the electronic component.

19. The method of claim 11, wherein the supporting legs cover side surfaces of the carrier structure.

20. The method of claim 11, further comprising clamping the supporting legs and the carrier structure via clamps.
